

AYE
FIT HUB

• HIT • DAY 2

HIIT 101

The Science of Fat Burning in 20 Minutes

Day 2 of 30 | High-Intensity Interval Training

Awet G. Nway

Mechanical Engineer & Fitness Coach

- Science of EPOC & fat loss
- 20-min no-equipment circuit
- HIIT formats & rules

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01 | What Is HIIT?

High-Intensity Interval Training (HIIT) alternates short bursts of maximum effort with brief recovery periods. Instead of jogging for 60 minutes, you push hard for 20–40 seconds, rest, and repeat. The result: more fat burned in less time, with benefits lasting hours after you stop.

The key mechanism is **EPOC** — Excess Post-Exercise Oxygen Consumption. After intense effort, your body burns extra calories for up to 24 hours restoring oxygen levels, repairing muscle, and clearing metabolic byproducts. Steady-state cardio stops burning the moment you stop. HIIT keeps working.

02 | HIIT vs Steady-State Cardio

Metric	HIIT (20 min)	Steady Cardio (45 min)
Calories during	250–350 kcal	300–400 kcal
Calories after (EPOC)	100–200 kcal extra	~10 kcal
Total fat burned	Higher (24-hr window)	Lower (stops at end)
Muscle preservation	High	Low
Time required	20 minutes	45–60 minutes
Equipment needed	None	None

■ INFO — Heart Rate Zones

HIIT works in Zone 4–5 (80–95% max HR). Estimate max HR: 220 – your age. Example: age 30 → max HR 190 bpm → target 152–180 bpm during work intervals.

03 | The Four HIIT Formats

There are four main HIIT formats. Today you will use **Work:Rest**. As you progress through the 30 days you will encounter all four.

Format	Structure	Best For	Day Used
Work:Rest	30 sec on / 30 sec off	Beginners, building base	Days 2, 8, 15
Tabata	20 sec on / 10 sec off × 8	Fat loss, max effort	Day 9, 24
EMOM	X reps every minute on the minute	Strength + cardio	Day 15
AMRAP	As many rounds as possible	Fitness testing	Day 28

■■ WARNING — 3 Common HIIT Mistakes

1. Skipping the warm-up → risk of injury.
2. Not going hard enough during work intervals → no EPOC benefit.
3. Doing HIIT every day → overtraining, no recovery. Limit to 3–4 sessions/week.

04 | Day 2 Workout — 20-Minute Fat Burner

Format: 30 sec work / 30 sec rest | Rounds: 3 | Equipment: None | Time: ~20 min

WARM-UP (4 min — never skip)		
Exercise	Duration	Notes
March in place	60 sec	Pump arms, raise knees high
Arm circles	30 sec	Forward then backward
Hip circles	30 sec	Both directions
Leg swings	30 sec	Hold wall for balance if needed
Light jumping jacks	60 sec	Controlled, steady pace
Inchworm walk-out	30 sec	Hands to floor, walk out and back

MAIN CIRCUIT — 3 Rounds (30 sec work / 30 sec rest)			
#	Exercise	Modification	Reps R1 / R2 / R3
1	High Knees	March in place	___ / ___ / ___
2	Squat Jump	Bodyweight squat	___ / ___ / ___
3	Push-Up	Knee push-up	___ / ___ / ___
4	Skaters	Side step (no hop)	___ / ___ / ___
5	Burpee	Step-back burpee	___ / ___ / ___
6	Mountain Climbers	Slow mountain climbers	___ / ___ / ___

■ How Hard Should "Hard" Feel?

During work intervals: you should NOT be able to hold a full conversation. A word or two is fine. If you can talk freely — go faster. If you cannot breathe at all — slow down slightly. Aim for 8/10 effort.

COOL-DOWN (3 min)	
Exercise	Duration
Walk in place — bring heart rate down	60 sec
Standing hamstring stretch (each leg)	30 sec
Hip flexor lunge stretch (each side)	30 sec
Chest opener — arms wide, deep breath	30 sec
Child's pose or seated forward fold	30 sec

05 | HIIT Quick Reference Card

Rule	Guideline
Frequency	Max 3–4 HIIT sessions per week. Never two days in a row.
Duration	15–25 minutes is enough. More is not better with HIIT.
Intensity	80–95% max heart rate during work intervals.
Nutrition before	Light snack 60–90 min before. Avoid heavy meals.
Protein after	Within 30–45 min after training to support recovery.
When to skip	Sick, injured, or under 5 hrs sleep — do mobility instead.
Progress indicator	Should feel slightly easier each week at same intensity.

06 | Day 2 Self-Check Question

Complete this before moving to Day 3. Honest reflection accelerates progress.

"After completing today's HIIT session, what was your biggest challenge — physical (breathing, legs) or mental (wanting to stop)? What specific adjustment will you make in your next HIIT session?"

Your answer:

Next up — Day 3:
Core Anatomy — Why Your Core Is More Than Abs

Community & support:
[t.me/ayetechub](#)